Concours - Sports Tournament Management Database.

Group D3
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1. Understand and Complete the Description of the Case Study/Problem Domain:

1.1 Purpose:

The purpose of the **Sports Database Management System (SDMS)** is to provide a centralized and automated platform that streamlines the management of sports tournaments. It replaces manual record-keeping with an efficient system that integrates registration, scheduling, real-time updates, and analytics. By supporting multiple user roles—**administrators, coaches, referees, players, and spectators**—the system ensures secure, role-specific access and smooth coordination.

Key objectives include:

Centralized Data Management – Single source of truth for teams, players, venues, scores, and standings.

Role-Based Access – Secure access for administrators, coaches, referees, players, and spectators.

Data Integrity – Validation rules to prevent duplicate entries, scheduling conflicts, and invalid data.

Analytics & Reporting – Insights into player performance, team statistics, venue usage, and historical comparisons.

1.2 Scope:

The **Sports Database Management System (SDMS)** project aims to address the challenges faced by **administrators, coaches, referees, players, and spectators**. The system will include multiple user modes: **Administrator, Coach, Referee, Player, and Spectator**, each having role-specific permissions and access to relevant features. The scope of the project covers a wide range of functionalities to streamline tournament operations and enhance coordination among stakeholders.

1.3 Target Audience:

- **Players:** Use this database for the team/individual registration.For checking the schedule and their current leaderboard status.
- **Coaches:** For registering their team into intended sports tournaments and getting regular updates on team position as well as overall performance and general scoreboard.
- **Referees:** For getting updates on the match and keeping updates on all the teams and their past data as well as the rules specified for a certain sport.
- **Sponsors:** Keeps details of organizations funding tournaments or teams.
- **Team owner:** Buying and Selling of players and teams, updates on the allotted team budget and allotment of the equipment required.
- **Team Captain:** Data of all the players and team details , their health updates and team standings, allocated medical staff and ground staff to the team.
- **Administrators / Organizers** – To understand the system's capabilities for managing teams, matches, referees, venues, and results.
- **Project Team** – to design and implement a database.

1.4 Description:

This database allows administrators to register teams and players, generate match fixtures, assign referees and venues, update results in real-time, and maintain historical records of past tournaments. It supports different user roles such as admins, coaches, referees, and players, each with specific access permissions. By centralizing tournament information, the system ensures smooth coordination, avoids scheduling conflicts, reduces paperwork, and provides quick access to performance statistics and rankings. It can be scaled from small local competitions to large multi-sport tournaments, making it a reliable and adaptable solution for sports organizations, colleges, and event managers

Current Challenges:

Messy Records – Paper and spreadsheets often cause mistakes and confusion.

Difficult Scheduling – Making fair match fixtures and avoiding clashes takes too much time.

Scattered Data – Past records are hard to find and not well organized.

Limited Access – Coaches, referees, and players cannot easily get the information they need.

2.Document the Requirements Collection/ Fact Finding Phase:

2.1. Background Readings:

2.1.1. BITS Open Sports Meet (BOSM) – BITS Pilani

Major sports events include Athletics, Badminton, Basketball, Bodybuilding, Chess, Cricket, Football, Frisbee, Hockey, Volleyball, Squash, Swimming, Table Tennis, Tennis, Taekwondo, and Powerlifting, requiring large-scale scheduling and result tracking.

The organizational structure has **four main departments**: Controls, Sponsorship & Marketing, Publications & Correspondence, and Reception & Accommodation, each handling critical data flows.

Manual handling of **registrations, scheduling, venue allocation, score reporting, and logistics** creates inefficiencies and errors.

In addition to sports, BOSM has **social connect initiatives** such as *Junoon* (sports for specially abled children), *BOSM Marathon*, *YODH*, and *Play for a Cause*, which add further layers of participant and event management.

Hence, the database must also support **non-sport events**, NGO involvement, and cause-driven initiatives.
References :-https://en.wikipedia.org/wiki/BOSM

http://www.knowafest.com/2011/09/bosm-2011-bits-pilani-sports-festival.html

2.1.2. Udghosh – IIT Kanpur:-

Udghosh is the annual sports festival of IIT Kanpur, one of Asia's largest college sports meets with participation from 450+ colleges and 40+ organizations.

The fest hosts a **wide range of sports** such as athletics, basketball, cricket, football, hockey, volleyball, kho-kho, squash, badminton, table tennis, tennis, and even unique events like roller hockey, marathon, and adventure activities.

The fest is **organized entirely by students**, led by a core committee of **~23 members** (Overall Coordinators + Department Heads) with support from **800+ organizers and 2,500+ ambassadors,** making it one of the largest student-driven events in the country.

The **organizational structure** is divided into departments like Accounts, Sponsorship & Marketing, Hospitality, Media & Publicity, and Sports Coordination, each handling different responsibilities that require proper communication and tracking.

References :-https://www.iitk.ac.in/new/udghosh-21

https://x.com/iitkanpur/status/900221622665261056

2.1.3. Riviera – VIT Vellore

Annual international fest with **30,000+ participants from 650+ colleges**.

Includes traditional sports (basketball, football, tennis) and **unique events** like **Arm Wrestling (Guinness Record), Human Foosball, Soap Up, and E-Sports tournaments**.

Pro-shows by international artists, **literary contests** in multiple languages, and **social campaigns** with NGOs.

Involves handling **registrations, scheduling, accommodation, scores, sponsorships, and publicity**.

References :-Riviera

https://vit.ac.in/event/riviera-2025

2.1.4. Shaurya – IIT Kharagpur

Sports & Events: Shaurya is IIT Kharagpur's flagship sports festival, featuring **20+ sporting disciplines**, adventure sports, and unique attractions such as an **E-Sports Talent Arena** (BGMI, Valorant, FIFA), an **F1 race car simulator**, and a **virtual running challenge** called the *Shaurya Run*.

Beyond sports, the fest integrates **cultural shows, workshops, guest lectures, and carnival games**, making it a comprehensive platform for competition, entertainment, and skill development.

The fest is entirely **student-run**, led by a core committee and supported by a wide network of organizers and campus ambassadors. Its **departmental structure**—including Sponsorship & Marketing, Hospitality, Media & Publicity, and Sports Coordination—ensures efficient role distribution.

References *:* Shaurya IIT Kharagpur

2.1.5. Concours (DA-IICT, Gujarat)

Concours is the renowned **annual sports fest of DA-IICT**, offering a competitive platform for students from around India. It typically spans multiple days and is organized by the institute's Sports Committee.

Features core sports such as **badminton, basketball, chess, carrom, cricket, football, table tennis, lawn tennis, volleyball**, and a **marathon (Youth Run)**.

Includes playful and competitive variations like **3-on-3 volleyball, 3-point basketball shootouts, futsal, Gully Cricket, Anti-Chess, Human Tic-Tac-Toe, Mixed Badminton, Foot-Volley-Ball, Arm Wrestling, and Dart Games**.

Managed by the sport committee, ensuring participants are professionally accommodated and logistics are streamlined.

References :- https://www.youtube.com/watch?v=419lmT75AU8

http://concours.daiict.ac.in/

Combined Requirements from Background Readings

The system must allow smooth **participant and team registration**, categorizing entries by institute and event type while maintaining their history for future editions. It should support **event scheduling** through automated fixture generation and prevent clashes across multiple sports, cultural, and e-sport activities. A robust **venue and resource allocation** mechanism is needed to dynamically assign grounds, halls, and equipment with real-time updates.

The platform should provide **current result and score tracking**, leaderboards, and archival of past data. For finances, it must handle **sponsorship and expense management** with proper reporting and audits.

A structured **committee and role management** system is necessary to define access levels, assign tasks, and streamline coordination. The database should also support **unique and informal events**, manage **ticketing for cultural shows and VIP passes**, and facilitate **communication and publicity** through notifications and social media integration.

Non-functional needs include high **performance** for real-time updates, **scalability** for future events, **reliability** with backups, **user-friendly design**, **secure role-based access**.

Core Functional Requirements

The system must manage participants, teams, and institutes with event history, support

automated scheduling of events with conflict checks, ensure venue and resource allocation with real-time updates, and allow live result entry, leaderboards, and archival of records.

Administrative & Support Features

It should handle sponsorship and finance tracking with reports, provide logistics for accommodation, food, and transport, maintain committee hierarchy with role-based access and task tracking, and allow flexibility for unique or informal events.

2.2. Interview:

Interview Plan 1:

Date: 2025-09-01

Time: 4:00 PM

Duration: 25 minutes

Place: Nescafe Coffee Shop

Purpose of Interview:

Preliminary meeting to identify problems and requirements for developing a database management system for concours tournament.

Participants:

- Saloni Vaghela.
- Yashvi Patel.
- Madhav Kanjilimadom (Football Team Player).

Agenda:

- **Problems and Concerns:** Discuss current issues with tournament management.
- **Current Procedures:** Review how tournaments are currently organized, from registration to results.
- **Initial Ideas:** Brainstorm on what a new database system could offer.
- **Follow-up Actions:** Determine next steps and any necessary information to gather.

Interview Summary 1:

- **Current Process Issues:**
  The tournament is managed using manual paperwork and notice boards, with schedules sometimes shared informally via WhatsApp groups. This often leads to delays, miscommunication, and confusion for players.

- **Registration Challenges:**
  Players and teams are required to fill out lengthy google forms every year, repeating the same personal and team details. This is time-consuming and error-prone. A digital system could store reusable player profiles.

- **Scheduling & Updates:**
  Match fixtures are posted late, and last-minute changes are difficult to communicate. Players often miss updated information, causing teams to arrive late or unprepared. Real-time scheduling would solve this issue.

- **Match Results & Standings:**
  Results are manually recorded and updated much later on boards or spreadsheets. This creates uncertainty about rankings and upcoming fixtures. Football-specific stats (goals, assists, saves, cards) are not tracked properly.

- **Motivation & Records:**
  Currently the previous historical tournament data (team performance, individual player statistics across seasons) is not available for the players. As if it is there then it's a good way to encourage players to track progress and improve performance.

Interview Plan 2:

Date: 2025-08-31

Time: 10:00 AM

Duration: 20 minutes

Place: Nirma University.

Purpose of Interview:

Gather requirements and identify challenges from the coordinator's perspective for preparing the SRS of a Sports Tournament Database Management System.

Participants:

- Saloni Vaghela.
- Yashvi Patel.
- Krisha Vaghela (Coordinator of sports events at Nirma University).

Agenda:

- Discuss current challenges in managing tournament registrations, scheduling, and results.
- Understand expectations for an automated database system.

Interview Summary 2:

Current Process Management:
The coordinator explained that most tournament activities, such as team registrations, scheduling, and result tracking, are still handled manually using spreadsheets, paper forms, and notice boards. This often leads to errors, delays, and difficulties in managing large numbers of participants.

Registration Issues:
Teams from various colleges submit registration forms separately, often in inconsistent formats. Verifying eligibility, payment status, and participant details is time-consuming. Duplicate or incomplete entries are common.

Scheduling & Logistics:
Preparing fixtures across multiple sports, ensuring there are no time or venue clashes, and rescheduling due to weather or unforeseen events is a major challenge. Currently, these updates are managed manually and are hard to communicate to all stakeholders quickly.

Result Management:
Match scores and outcomes are reported manually by referees/umpires and then entered into spreadsheets. This creates delays in updating standings, and errors sometimes go unnoticed. The lack of real-time updates frustrates both players and spectators.

Desired Features for Organizers:
The coordinator emphasized the need for a centralized dashboard where organizers can manage registrations, schedules, venues, officials, and results in one place. Automated notifications for participants and live updates for spectators would improve overall efficiency and engagement.

Record Keeping & Reports:
Organizers would benefit from a system that generates automated reports – such as participation summaries, disciplinary records, team statistics, and final standings – which are often required for archives and post-tournament analysis.

Supporting Documents:
The coordinator offered to provide past tournament registration forms, fixture spreadsheets, result sheets, and volunteer duty rosters to guide requirement specification.

Interview Plan 3:

Date: 2025-08-30

Time: 2:00 PM

Duration: 15 minutes

Place: SAC-1

Purpose of Interview: Gather insights from the Sports Committee's perspective for the SRS of the Sports Tournament Database Management System.

Participants:

- Saloni Vaghela.
- Yashvi Patel.
- Rishita Jodhawat (Sports committee member at DAU)

Agenda:

- Discuss current issues in managing registrations, scheduling, results, and communication.
- Review how tournaments are presently organized and maintained.
- Gather requirements for a centralized database system to improve efficiency and record-keeping.

Interview 3 Summary:

Manual Dependency:
Most tournament activities like registrations, scheduling, and result tracking rely heavily on paperwork and spreadsheets, which is time-consuming and error-prone.

Data Inconsistencies:
Repeated collection of the same player/team details each year causes redundancy and mistakes, making registration inefficient.

Scheduling Bottlenecks:
Fixtures are created manually and any changes due to delays or venue issues are difficult to communicate effectively to all participants.

Delayed Updates:
Results are recorded on paper first and then updated later, which slows down the publishing of standings and sometimes leads to inaccuracies.

Communication Gaps:
Coordination among committee members, referees, and players happens through fragmented channels like calls and WhatsApp, often causing miscommunication.

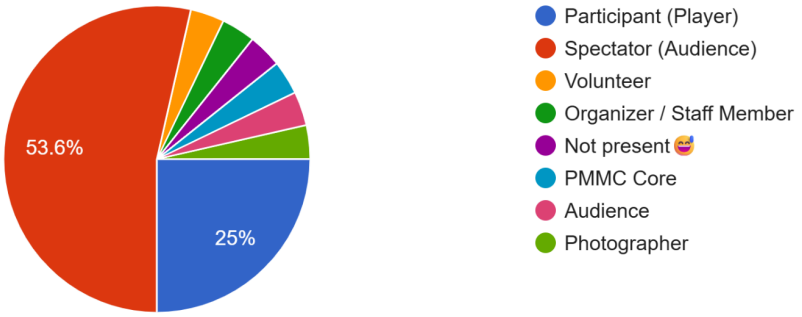
Record Management Issues:
Tournament history and performance statistics are not stored in a centralized way, making it hard to track past participation and achievements.

System Vision:
A new system should streamline operations, reduce manual work, ensure accurate real-time updates, and provide centralized access for all stakeholders.

2.3. Questionnaire:

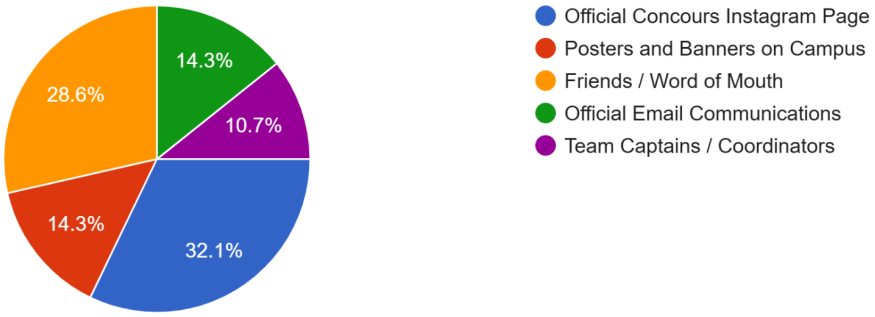
What was your primary role during Concours 2024?

28 responses



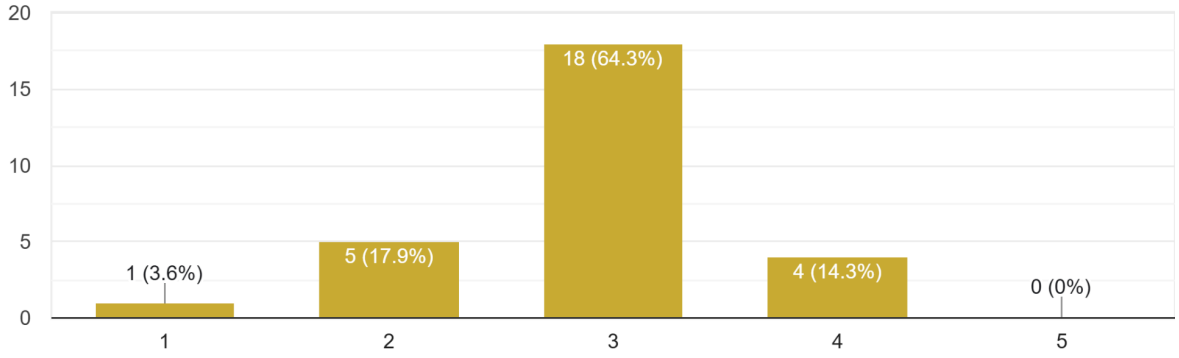
How did you receive most of your information about

28 responses



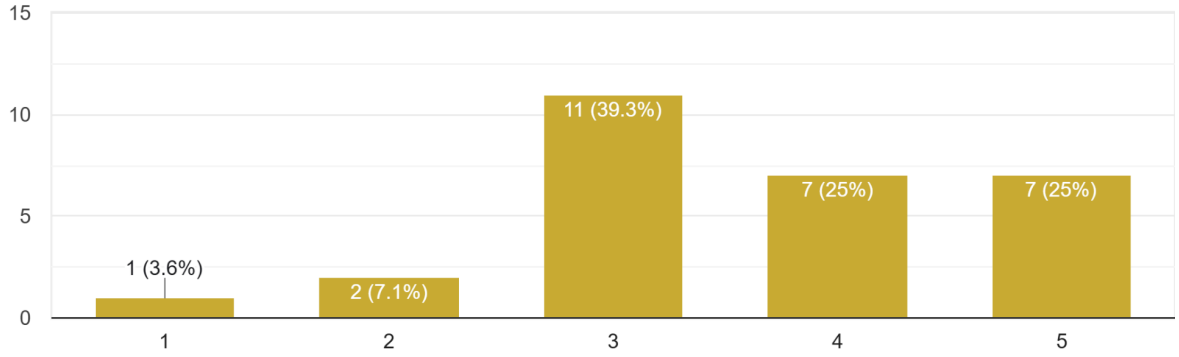
On a scale of 1 to 5, how easy was it to find information about specific match timings and venue locations? (1=easy...5=too hard)

28 responses



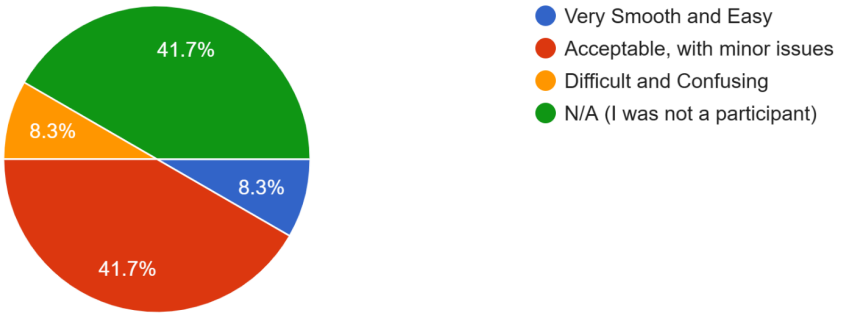
How interested would you be in a feature that lets you browse results, winners (like Ahmedabad University's House Cup win), and records from previous year's Concours events?

28 responses

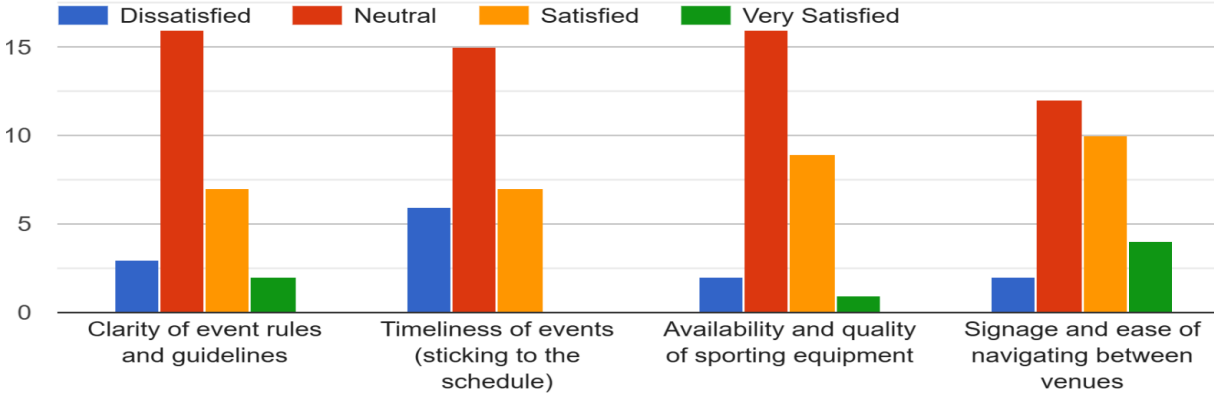


If you were a participant, how would you rate the registration process for your event(s)?

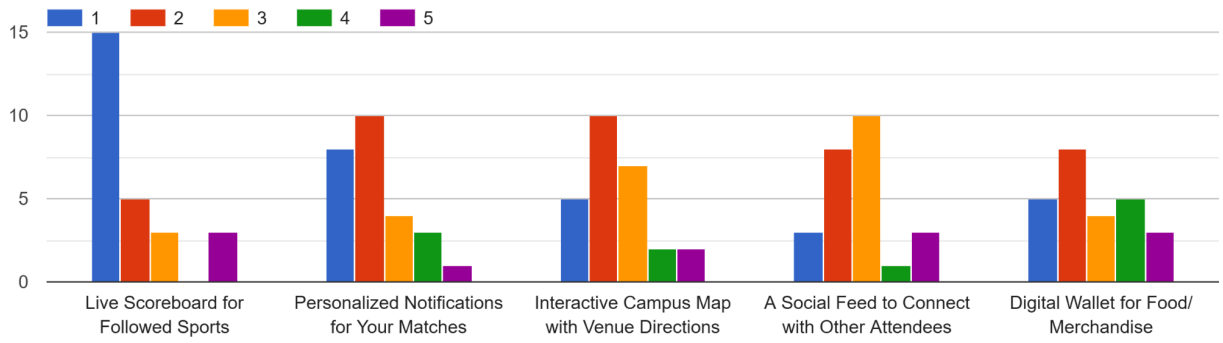
24 responses



Please rate your satisfaction with the following logistical aspects of Concours 2

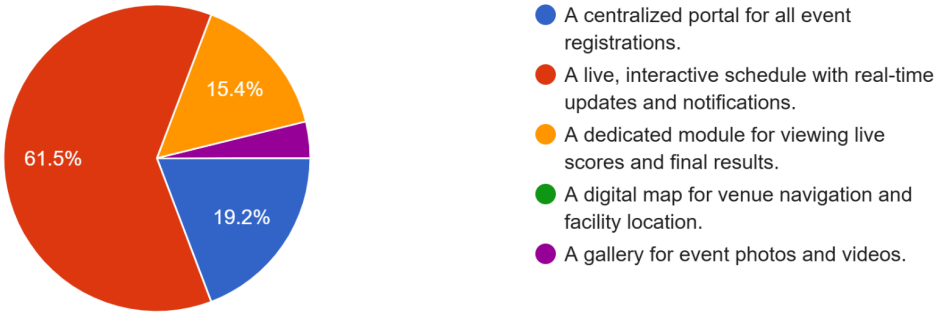


10. Imagine a future Concours app with a personal dashboard. Please rank the following features from most to least important. (1=Most Important)



For our Event Management System, which ONE feature would add the most value to your experience?

26 responses



2.4. Observations:

- **Fragmented Communication:** Information is currently scattered across multiple channels, with word of mouth, and the Official Instagram Page being the primary sources. This is unreliable and inefficient.
    - **System Solution:** The system must act as a **centralized information hub**, eliminating the need for attendees to check multiple sources.

- **Poor Schedule Management:** This is the most significant pain point. **41.7%** of respondents cited **"unclear or frequently changing schedules"** as a major issue. Open-ended comments explicitly demand better "time management" and for events to "happen at the time stated."
    - **System Solution:** The database must support a **live, interactive schedule**. It needs to be easily updatable by organizers, with any changes (delays, venue swaps) instantly pushed as **personalized notifications** to relevant participants and followers.

- **Lack of Real-Time Data:** Attendees, particularly the **53.6% who are spectators**, are disconnected from the action. **37.5%** felt the lack of real-time scores negatively impacted their experience.
    - **System Solution:** A **live scoreboard** was ranked as the **most important feature** for a future app. The database schema must be designed to handle real-time score entry, calculate standings, and display detailed statistics beyond just the final score (e.g., player stats, MVPs, team leaderboards)

- **Logistical Inefficiencies:** Major issues like **"Overcrowding at popular matches" (66.7%)** and **"Difficulty finding specific venues" (20.8%)** detract from the experience. While finding information was rated a neutral **3 out of 5** in difficulty by most (64.3%), this is still far from an ideal, easy experience.
    - **System Solution:** The system can help manage this by incorporating an **interactive campus map** with venue directions (a highly-ranked feature). It could also potentially provide real-time crowd estimates for popular venues.

3. Fact-Finding Chart:

Objective	Technique	Subject	Time Commitment
To get an overview of the problem statement as well as the purpose if the database	Online data-lookup and various already existing resources.	Online Research, other similar projects on GitHub repos.	0.5 day
To establish scope and perspective for the system	Observation	2 Team Members	0.5 day
To establish an overview of how this system works for various other institutes.	Background Readings	2 Team Members	2 hours
To establish objectives of the new system	Interview	Current Team player. (Interview 1)	2 hours (interview+report)
To get an overview of the method followed by other institutes for their Sports Tournaments	Interview	Event Coordinator at Nirma University. (Interview 2)	1.5 hours (interview+report)
To establish additional functionalities/features for our system	Questionnaire	Target User as well as Audience	1 day
To know about the current management of the event and the challenges that are faced by the management team.	Interview	Event Coordinator at DAU(interview 3)	2 hours(interview + report)
To gather conclusion from the background readings as well as Questionnaire and interviews	Data Lookup	2 Team members	3 hours

4. Requirements:

1. Functional Requirements:

- Role-based access control (admin, referee, player, spectator).
- Tournament creation with details like name, type, rules, venue, and dates.
- Support for different tournament formats (knockout, league, round-robin).
- Automatic or manual fixture/schedule generation.
- Rescheduling option with notifications to participants.
- Registration of teams and players with personal and contact details.
- Management of team rosters (adding/removing players).
- Validation of player eligibility (age, department, course, etc.).
- Recording of match results, scores, and referee remarks.
- Automatic update of standings, points, and brackets after matches.
- Storage of match history and performance stats.
- Leaderboards for teams and individual players.
- Generation of reports (fixtures, results, rankings, player stats).
- Notifications for upcoming matches, results, and schedule changes (via app/email/SMS).
- Public access for spectators to view schedules, results, and leaderboards.
- Centralized database storing tournaments, teams, players, and results.
- Backup and restore support for database safety.

2. Non-Functional Requirements:

- Fast response time (within 2–3 seconds for queries).
- Secure login with encrypted passwords.
- Role-based data protection.
- Reliable uptime during tournaments.
- User-friendly interface for both web and mobile access.
- Support for running multiple tournaments simultaneously.
- Easy maintenance and future upgrades.
- Proper system documentation for admins.

3. System Constraints:

- Use of a relational database (e.g., MySQL, PostgreSQL).
- Accessible on standard college lab computers and mobile devices.
- Integration with email/SMS APIs for notifications.
- Cloud or on-premise deployment depending on college resources.

5. User Classes and Characteristics:

1. Administrators (System/Organizers)

- **Role:** Overall management of tournaments and system configuration.
- **Responsibilities:** Create tournaments, define rules, manage schedules, oversee registration, assign referees, manage database backup/restore.
- **Characteristics:**
    - Moderate technical knowledge.
    - Need secure, full access to all modules.
    - Require detailed reports and analytics.

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2. Referees/Officials

- **Role:** Enter and validate match results, fouls, penalties, and other game statistics.
- **Responsibilities:** Record scores, update results, confirm player eligibility.
- **Characteristics:**
    - Basic computer/mobile literacy.
    - Need a simple, fast interface for real-time updates.
    - Limited access (only to match data they officiate).

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3. Team Managers / Captains

- **Role:** Register and manage their team and players.
- **Responsibilities:** Submit player details, update roster, confirm participation, view schedules.
- **Characteristics:**
    - Limited technical background.
    - Require easy-to-use registration forms.
    - Access restricted to their own team data.

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4. Players

- **Role:** Participate in tournaments and view personal/team performance.
- **Responsibilities:** Register individually or through a team, check match timings, track statistics.
- **Characteristics:**
    - Mostly students with varying technical skills.
    - Access limited to personal profile, schedules, and results.

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5. Spectators (Students/Faculty)

- **Role:** View tournament details, schedules, results, and leaderboards.
- **Responsibilities:** Follow matches and rankings.
- **Characteristics:**
    - General users with no login required (read-only access).
    - Need fast, user-friendly access to live results.

6. Operating Environment:

6.1 Hardware Requirements:

- Any PC/laptop available.
- Mobile devices (Android/iOS smartphones, tablets) for referees, players, and spectators.
- Optional projector/display screens for live scoreboards.

6.2 Software Requirements:

- **Operating Systems:**
  - Server: Linux (Ubuntu/CentOS) or Windows Server.
  - Client: Windows 10/11, macOS, Android, iOS.
- **Database Management System:** MySQL or PostgreSQL (relational DB).
- **Third-party :** Google forms.

6.3 Network Requirements:

- College LAN for admin and referee access.
- Secure internet connectivity for remote access (e.g., players/spectators outside campus).
- Wireless access for mobile devices during tournaments.
- HTTPS protocol for secure communication.

7. Product Functions:

A) User Management

- Registration and login for administrators, referees, team managers, and players.
- Role-based access control with different privileges.

B) Tournament Management:

- Creation and configuration of tournaments (type, rules, venue, schedule).
- Automatic or manual generation of fixtures/brackets.
- Rescheduling and cancellation of matches with notifications.

C) Team & Player Management

- Registration of teams and individual players.
- Management of team rosters (add/remove players).
- Validation of eligibility (age, department, year of study, etc.).

D) Match Management

- Recording of match details (scores, referee remarks, fouls, penalties).
- Real-time update of tournament standings and brackets.
- Storage of match history and results.

E) Statistics & Reports

- Leaderboards for teams and players.
- Player and team performance statistics.
- Generation of reports in printable/exportable formats (PDF, Excel, CSV).

F) Communication & Notifications

- Alerts for upcoming matches, results, and schedule changes.
- Notifications via email/SMS/web portal.

G) Spectator Access

- Public viewing of schedules, live scores, and results.
- Access to leaderboards and tournament summaries.

H) Database Operations

- Centralized storage of tournaments, teams, players, and match data.
- Support for CRUD operations (create, read, update, delete).
- Backup and restore of database.

8. Privileges:

Access Control Matrix:

Function/Action	Administrator	Referee	Team Manager	Player
Create/Edit/Delete Tournament	Yes	No	No	No
Define Rules, Venues, Schedules	Yes	No	No	No
Approve/Reject Team Registrations	Yes	No	No	No
Register Team/Players	No	No	Yes	No
Update Team Roster	No	No	Yes	No
View Tournament Schedule	Yes	Yes	Yes	Yes
Enter Match Results/Scores	No	Yes	No	No
Validate Player Eligibility	Yes	Yes	No	No
View Match Results	Yes	Yes	Yes	Yes
Generate/Export Reports	Yes	No	No	No
Manage Database Backup/Restore	Yes	No	No	No
View Personal Profile and Stats	Yes	Yes	Yes	Yes
Receive Notifications	Yes	Yes	Yes	Yes
Access Live Scores/LeaderBoards.	Yes	Yes	Yes	Yes

9. Assumptions:

- All users (admins, referees, players, team managers) have access to a computer or smartphone with internet connectivity.
- Users have basic knowledge of using web or mobile applications.
- The college will provide and maintain the required hardware (servers, computers, network).
- Internet connectivity within the campus will be stable during tournaments.
- Players and teams will provide accurate details during registration (e.g., name, age, department, eligibility).
- Referees will enter match results promptly and accurately.
- The database will not exceed the expected size for a typical college-level tournament (few hundred players/teams).
- Only authorized users (admins/organizers) will have permission to modify tournament settings.
- The college IT staff will handle regular backups and maintenance of the system.
- External APIs (for email/SMS notifications) will remain available and functional.

10. Business Constraints:

- The system must comply with college policies and rules regarding student participation in sports.
- Only currently enrolled students and approved staff members are eligible to register in the system.
- A player may only represent one team per tournament (no duplicate participation).
- Teams must be registered before the official registration deadline; late entries will not be accepted.
- Match schedules must not clash with academic schedules (classes/exams).
- Data privacy of students (personal info, contact details) must be maintained as per college regulations.
- Tournament formats (knockout, league, etc.) are fixed as per college sports committee decisions.
- Budget limitations restrict the use of expensive third-party software; preference is for open-source or college-provided solutions.
- The system must be operable on existing college IT infrastructure (labs, servers, internet).
- Tournament data must be preserved for a defined period (e.g., 3–5 years) for record-keeping.
- Only authorized organizers can modify or cancel tournaments once they are created.
- Sponsorship/advertising (if any) must follow college and university guidelines.

Description:

Our database is for managing sports tournaments held at university level that helps in keeping all the scheduling, participant data , scores and leaderboards.

It replaces manual record-keeping with an efficient system that integrates registration, scheduling,and analytics.

It offers role-specific access to various intended audiences, that is, administrators, referees, coaches, players and spectators (audience).

Intended Users:

1) Player: Players can access the schedule of the matches and check their team details as well as their own profile in the database.Check recent statistics as well the previous season statistics.

2) Captain: Captains can access the data of all players and check team standings as well as check the recent and previous season performance data of the team.

3) Referees: Can access the data of all players and teams.

4) Audience : Can access team-details, that is list of players,their coach and captains, schedule of the matches and location details.

5) Coach: Can access all the details of the team i.e, individual player details as well as team standings and overall leader-board, referees allotted to the match and schedule and location details.

Entity	Attributes
Player	player_id, name, age, joining_year
Captain	captain_id, name, age, team_id, player_id, sport_id, joining_year
Coach	coach_id, name, age, team_id, joining_year, sport_id
Sport	sport_id, sport_name, type (Team/Individual)
Venue	venue_id, name, location
Team	team_id, name, sport_id, college_name
Matches	match_id, sport_id, venue_id, time, status (Live/Post)
Result	result_id, match_id, team1_score, team2_score
Tournament	tournament_id, sport_id, sport_name, start_date, end_date, organizer, location
Team_Statistics	team_id, tournament_id, match_id, standings
Player_Statistics	player_id, match_id, sport_id, team_id, tournament_id, standings
Leaderboard	team_id, match_id, tournament_id, standings, sport_id
Recent_Stats	team_id, tournament_id, standings, match_id, sport_id
Health	player_id, age, weight, height, blood_group
Equipment	equipment, number_of_equipment, sport_id, sport_name
Coordinator	coordinator_id, name, sport_id, tournament_id, match_id

Key Insights from Background Reading & Data Lookup

- **Diverse Event Support**: The system must be flexible enough to handle a wide range of events beyond traditional sports, including unique variations (e.g., Gully Cricket, Human Foosball), E-Sports, cultural shows, and social initiatives.

- Manual processes lead to **scheduling clashes, delayed results, and loss of historical records**.

- **Scalability**: The database needs to manage large volumes of data from thousands of participants across hundreds of colleges, similar to the scale of fests like Udghosh and Riviera.

- **Committee and Role Management**: A structured system is needed to manage the student-run organizing committees by defining roles, assigning tasks, and tracking responsibilities to ensure smooth coordination, as seen in the departmental structures of BOSM and Udghosh.

- **Data Archiving**: The ability to maintain a history of participants, teams, scores, and winners from previous years is essential for tracking records and engaging returning participants.

- **Centralized Communication**: The platform must act as a single source of truth to streamline communication and publicity, integrating notifications and social media updates to keep all stakeholders informed.

Functionalities to be included in the database are:

- **User and Role Management**: The system needs distinct access levels and privileges for different user classes such as Administrators, Referees, Team Managers/Captains, Players, and Spectators.

- **Tournament and Schedule Management**: Administrators must be able to create and manage tournaments, define rules, generate match fixtures (supporting formats like knockout and league), and reschedule events with automated notifications to participants.

- **Team and Player Registration**: The system should streamline the registration of teams and players, manage team rosters, and validate player eligibility based on criteria like age and course enrollment.

- **Results and Statistics**: Referees will need a simple interface to record match scores and other game statistics . This data will automatically update standings, leaderboards, and historical performance stats for both teams and individual players.

- **Spectator Access**: A public, read-only interface is required for spectators to view schedules, live scores, and tournament leaderboards without needing to log in.